A comparative study of hybrid approaches for solving capacitated lot sizing problem with setup carryover and backordering

Hacer Guner Goren*

Department of Industrial Engineering,
Pamukkale University,
Kinikli Campus, 20070, Denizli, Turkey
Email: hgoren@pau.edu.tr
*Corresponding author

Semra Tunali

Department of Business Administration,
Izmir University of Economics,
Sakarya Caddesi,
No. 156, 35330, Balcova-Izmir, Turkey
Email: semra.tunali@izmirekonomi.edu.tr

Abstract: The classical capacitated lot sizing problem is shown to be NP-hard for even a single item problem. This study deals with an extended version of this problem with setup carryover and backordering. To solve this computationally difficult lot sizing problem, we propose a number of hybrid meta-heuristic approaches consisting of genetic algorithms and a mixed integer programming-based heuristic. This MIP-based heuristic is combined with two types of decomposition schemes (i.e., product and time decomposition) to generate subproblems. Computational experiments are carried out on various problem sizes. We found that hybrid approaches employing only time decomposition scheme or combination of both time and product decomposition schemes in different forms outperform the other hybrid approaches. Moreover, we investigated the sensitivity of the two best performing approaches to changes in problem-specific parameters including backorder costs, setup times, setup costs, capacity utilisation and demand variability. [Received 2 November 2015; Revised 25 March 2016; Accepted 7 April 2016]

Keywords: lot sizing; setup carryover; backordering; hybrid approaches; sensitivity analysis; paired-t test.

Reference to this paper should be made as follows: Goren, H.G. and Tunali, S. (2016) ‘A comparative study of hybrid approaches for solving capacitated lot sizing problem with setup carryover and backordering’, *European J. Industrial Engineering*, Vol. 10, No. 6, pp.683–702.

Biographical notes: Hacer Guner Goren is an Assistant Professor at the Department of Industrial Engineering, Pamukkale University, Turkey. She received her BS in Industrial Engineering from Dokuz Eylul University, Turkey in 2002, MS in Industrial Engineering from Pamukkale University, Turkey in 2005 and PhD in Industrial Engineering from Dokuz Eylul

University, Turkey in 2011. During her PhD, she had been a visiting student at HEC Montreal for one year. Her main research interests are modelling algorithms for production planning problems and multi criteria decision making.

Semra Tunali received her BS in Industrial Engineering in 1978, MS in Applied Statistics in 1980 from Ege University, Turkey, MBA from University of Missouri, USA in 1985 and PhD in Computer Science from Ege University, Turkey in 1991. Before joining Izmir University of Economics in 2010, she worked at Department of Industrial Engineering, Dokuz Eylul University, Turkey. She has been as a Visiting Professor at GINTIC Institute of Manufacturing Technology in Singapore, Northwestern University in Chicago, USA, and Loyola University in Chicago, USA. Her main research interest is performance improvement of manufacturing systems using simulation modelling and meta-heuristics.

1 Introduction

In today's competitive world, satisfying customer requirements on time is the main goal of every organisation. However, sometimes it may not be possible to achieve this goal due to uncertainties in lead times, order processing times and unexpected breakdowns in the production line, etc. In these cases, either stockouts are lost or they are backordered. This study suggests hybrid meta-heuristic approaches to solve capacitated lot sizing problem (CLSP) with backorder option that allows to meet unsatisfied customer demand at a later period. Lot sizing problems try to answer two main questions. The first one is related to determining the periods where production should take place. The second one is about the quantities that should be produced in these periods while satisfying customer requirements with the possible minimum cost. These problems can be classified into two groups according to the type of demand. Static lot sizing problems consider static demand whereas dynamic lot sizing problems focus on dynamic demand. The well-known CLSP deals with capacity constraints under dynamic demand. The CLSP problem can be extended by adding setup times, setup carryover, sequence-dependent setup costs, backordering, parallel machines, etc. This study, in addition to allowing backorders, it incorporates setup carryover to solve CLSP ($CLSP^+$). The setup carryover allows carrying at most one setup state of a product from one period to another and if demand for a product can not be satisfied on time in a period, it can be backordered. Knowing that the general case of the single-item CLSP is NP-Hard (Florian et al., 1980), $CLSP^+$ considering both setup carryover and backordering is computationally even more difficult. Although the literature focusing on the CLSP with setup carryover is rather rich (Sox and Gao, 1999; Suerie and Stadtler, 2003; Gopalakrishnan et al., 2001; Almada-Lobo et al., 2007; Goren et al., 2012; Goren and Tunali, 2015), the $CLSP^+$ has not attracted the attention of many researchers. Only a few studies focusing on this problem have been noted during the literature survey. In one of these, a four-stage greedy heuristic is proposed by Karimi and Ghomi (2002) to solve $CLSP^+$. In another study, Karimi et al. (2006) propose a tabu search (TS)-based approach and use this four-stage greedy heuristic to start the search with a feasible initial solution. A new solution procedure is proposed in Quadt and Kuhn (2009) to solve $CLSP^+$ in a parallel machine environment motivated by a real life problem.

Considering the features of the problem, commercial solvers are not able to solve large size lot sizing problems in reasonable computational time. Therefore, heuristic approaches are useful in this case. A review of current literature on applications of GAs to solve lot sizing problems (Goren et al., 2010) reveals that using GAs to solve $CLSP^+$ has not attracted any researcher. In this study we propose GA-based hybrid approaches to fill this gap. It is well known that pure GAs can quickly identify promising areas in the search space, but they are not good at reaching the optimum in large complex search spaces. To improve the performance of GAs for solving this computationally difficult problem we hybridise GAs with a MIP-based heuristic.

The remainder of the study is organised as follows. In the following section, the $CLSP^+$ is defined. The proposed hybrid approaches to solve this problem are presented in section three. The fourth section presents the computational studies including the results of paired-t test checking whether the differences between the best performing proposed hybrid approaches are statistically significant and investigation of the robustness of these approaches under problem specific parameters. Finally, the last section summarises the findings of this study and presents some future research directions.

2 Problem definition

The CLSP is the extension of Wagner-Whitin (1958) problem which focuses on only one product and ignores capacity constraints. Different from Wagner-Whitin problem, more than one product can be produced in one period under capacity constraints in the CLSPs. The mathematical model of the CSLP has been proposed by Manne (1958). The main logic of the problem is to produce n products on one machine. The parameters and decision variables are explained below.

Decision variables:

X_{it} production quantity of product i in period t

Y_{it} 1, if there is setup for product i in period t

0, otherwise

I_{it} inventory level for product i in period t

Parameters:

s_{it} setup cost for product i in period t

C_t capacity of the resource in period t

h_{it} holding cost of one unit of product i in period t

p_{it} production cost of one unit of product i in period t

d_{it} demand of product i in period t

d_{itk} cumulative demand of product i for period t until k

a_i production time of one unit of product i

K number of products

P number of periods

$$Min\sum_{i\in K}\sum_{t\in P}(s_{it}Y_{it}+p_{it}X_{it}+h_{it}I_{it}) \tag{1}$$

s.t.

$$I_{it-1}+X_{it}-I_{it}=d_{it} \quad \forall i\in K,t\in P \tag{2}$$

$$X_{it}\leq \min\{C_t/a_i, s\,\mathrm{d}_{itm}\}Y_{it} \ \forall i\in K,t\in P \tag{3}$$

$$\sum_{i\in K}X_{it}a_i\leq C_t \quad t\in P \tag{4}$$

$$X_{it},I_{it}\geq 0; Y_{it}\in\{0,1\} \quad \forall i\in K,t\in P \tag{5}$$

The objective function minimises the total cost which is composed of setup, production and holding costs. The second constraint is the demand constraint, where all customer demand should be satisfied. The upper bound on production of each product is satisfied with the third equation. The production quantity that can be produced in each period is limited to the capacity in each period and the cumulative demand from period t until period m. The fourth constraint is the capacity constraints. The fifth constraints are the non-negativity and binary constraints. By adding setup times, the classical CLSP can be extended. In that case, the fourth constraints should be revised as in the following. It should be noted that st_i shows the setup time for product i in constraint (6).

$$\sum_{i\in K}X_{it}a_i+st_iY_{it}\leq C_t \quad t\in P,i\in K \tag{6}$$

The literature focusing on the CLSP with setup times is rather rich (Manne, 1958; Trigeiro et al., 1989; Gopalakrishnan et al., 2001; Degraeve and Jans, 2007; Hindi et al., 2003; Jans and Degraeve, 2004). This paper focuses on the CLSP with setup carryover and backordering. To show the problem mathematically, the mathematical model given above should be revised as in the following (Suerie and Stadtler, 2003). First, new decision variables should be defined.

U_{it} a binary variable showing the setup carryover variable. If a setup state for product i is carried from period $(t-1)$ to (t), U_{it} is equal to 1.

Q_t the single product variable which equals one if the resource is occupied solely by product i in period t and is a binary variable.

I_{it}^{+} inventory level for product i at the end of period t.

I_{it}^{-} backorder level for product i at the end of period t.

Parameter:

b_i the backordering cost of product i.

The new mathematical model is given in the following.

$$Min\sum_{i=1}^{K}\sum_{t=1}^{P}s_{it}Y_{it}+\sum_{i=1}^{K}\sum_{t=1}^{P}h_{it}I^{+}_{it}+b_iI_{it}^{-} \tag{7}$$

Subject to

$$I^{+}_{i,t-1} + X_{it} - I^{+}_{it} - I^{-}_{i,t-1} + I^{-}_{it} = d_{it} \quad \forall i \in K; \forall t \in P \tag{8}$$

$$\sum_{i=1}^{K} X_{it} a_i + \sum_{i=1}^{K} Y_{it} st_i \le C_t \quad \forall t \tag{9}$$

$$\sum_{i=1}^{K} U_{it} \le 1 \quad \forall t \tag{10}$$

$$U_{it} \le U_{i(t-1)} + Y_{i(t-1)} \quad \forall i, t \tag{11}$$

$$U_{i(t+1)} + U_{it} \le 1 + Q_t \quad \forall i, t \tag{12}$$

$$Y_{it} + Q_t \le 1 \quad \forall i, t \tag{13}$$

$$X_{it} \le M_{it}(U_{it} + Y_{it}) \quad \forall i, t \tag{14}$$

$$Q_t \ge 0 \quad \forall t \in \{1, ..., P-1\} \tag{15}$$

$$U_{it} \in \{0,1\} \quad (U_{i1} = 0), Y_{it} \in \{0,1\} \quad \forall i \in K, \forall t \in P \tag{16}$$

$$X_{it}, \quad I^{+}_{it} \ge 0, \quad I^{-}_{it} \ge 0 \quad \forall i \in K; \forall t \in P \tag{17}$$

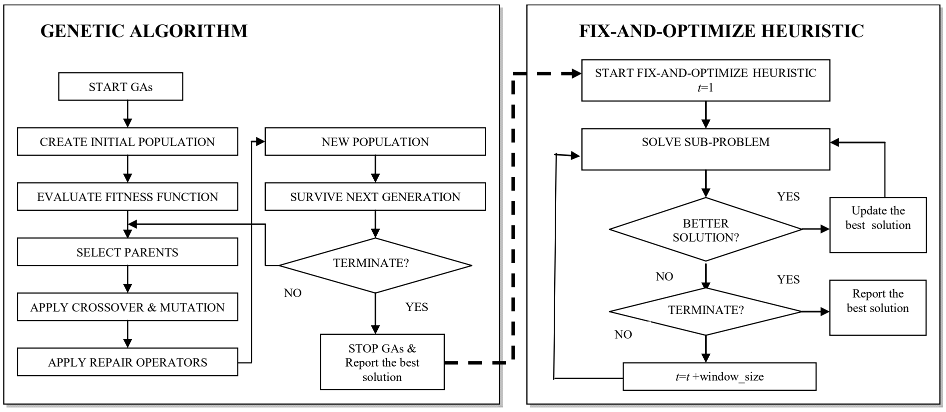
$$I^{-}_{it} = 0, \quad \forall i \in K; t = P \tag{18}$$

The total inventory, backorder and setup costs are minimised in the objective function (7). Equation (8) shows the inventory balance which shows the demand is satisfied either by production, inventory or backorder. The capacity constraints considering both setup and production times for each period are shown in (9). Constraint (10) states that at most, one setup state can be carried over on a resource from one period t to the next. A setup can be carried over to period t only if either the item is setup in the previous period or if the setup state is already carried from period (t – 2) to (t – 1) as stated in constraint (11). A setup state for a specific item can only be preserved over two bucket boundaries if that product is the only one produced in the middle period [constraint (12)], which is only possible if there is no setup in the middle period [constraint (13)]. If we have strictly positive production for a specific item, constraint (14) enforces either a setup or a carryover. The condition that there are no setup carryovers in the first period are stated in constraint (16). The non-negativity and binary conditions for the decision variables are shown in constraints (15) to (17). Lastly, constraint (18) shows that all demand must be satisfied during the planning horizon.

3 Proposed hybrid approaches

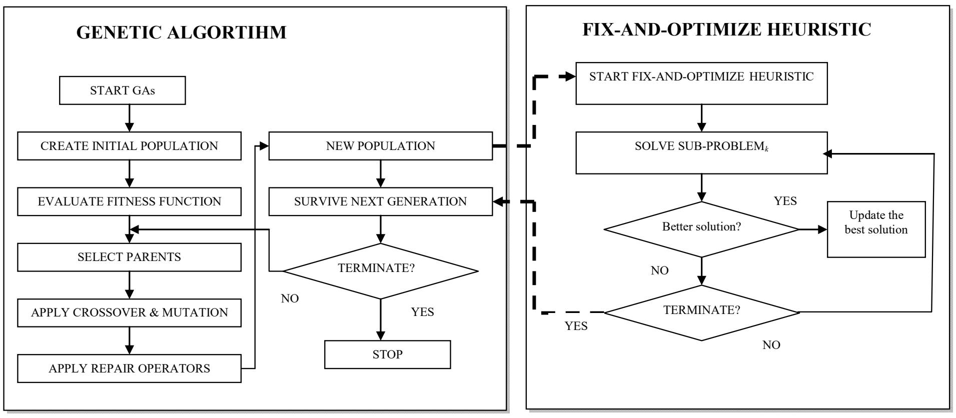
The single item CLSP is shown to be NP-Hard by Florian et al. (1980) and Bitran and Yanasse (1982). When other constraints such as setup times, parallel machines, backordering, setup carryover, etc., are introduced in the model, the problem becomes even more complicated. Therefore, it is difficult to find solutions to this problem with exact solution approaches in reasonable computational time. In recent years, heuristic methods have proven to be successful in solving many combinatorial optimisation problems. Inspired by this trend, in this study, we propose two groups of hybrid meta-heuristic approaches, which combine genetic algorithms (GAs) and a mixed integer programming (MIP)-based heuristic, namely fix-and-optimise (FOPT) heuristic to solve CLSP$^+$. After running the GAs for a pre-determined number of generations, the approaches of the first group, named sequential hybrid approaches, employ FOPT on the best solution obtained from GAs (see Figure 1).

Figure 1 The control logic of the proposed sequential hybrid approaches



To enable that the FOPT acts like a diversification strategy and guides GAs (see Figure 2), in the second group approaches, named embedded hybrid approaches, the FOPT is embedded into the loop of GAs.

Figure 2 The control logic of the proposed embedded hybrid approaches



The elements of the proposed approaches are summarised briefly in the following.

3.1 Fix-and-optimise heuristic

The fix-and-optimise heuristic (FOPT) is based on decomposition of the main problem which leads to a number of MIP subproblems. In each iteration of this MIP-based heuristic, a subproblem with small number of binary variables, i.e., most of the binary setup and carryover variables are set to fixed values, is solved using a commercial MIP solver. In recent years, the number of applications of FOPT heuristic in lot sizing problems have increased. In some applications, FOPT heuristic is used as a stand-alone heuristic (Sahling et al., 2009; Lang and Shen, 2011; Xiao et al., 2013) in others, it is hybridised with other heuristic approaches for performance improvement (Seanner et al., 2013; Toledo et al., 2013). It should be noted that our earlier study, Goren et al. (2012) is the first one combining FOPT heuristic with GAs in lot sizing. For further information on the FOPT heuristic, the reader can refer to Pochet and Wolsey (2000), Sahling et al. (2009) and Helber and Sahling (2010). As explained below, in this study, the subproblems are created using two decomposition schemes.

3.1.1 Product decomposition

To form the sub-problems, this decomposition scheme focuses on products instead of time periods. Particularly, based on some product specific information such as setup costs and holding costs, first a priority is assigned to each product, and sub-problems are ordered according to these priorities. Starting with high priority product the relevant binary setup and setup carryover variables are optimised while the rest is set to the fixed values based on the best solution obtained so far. In this study, five criteria are used to give a priority to the products: the setup cost, the holding cost, the ratio of setup cost to holding cost, the total demand of a product over the planning horizon and the total cost. The total cost is approximated through the economic order quantity (EOQ) model using the equation $\sqrt{2A_i h_i \overline{d}_i}$, where $\overline{d}_i$ is the average demand of product i over the planning horizon and A_i, h_i are the setup and holding costs of product i, respectively. The results of a computational study investigating the effects of these five criteria on performance of the proposed product decomposition scheme are given in section four.

3.1.2 Time decomposition

To form subproblems, this decomposition scheme focuses on time periods instead of products. The whole planning horizon is divided into time windows and for each time window a subproblem is formed. Starting from the first period, the binary setup and setup carryover variables are optimised in each time window. As in Goren et al. (2012), each sub-problem is defined for a time window of five consecutive periods with respect to the setup and setup carryover variables. For example, in one iteration (l) of the fix-and-optimise heuristic, three sub-problems are solved for an instance with 15 periods. The sub-problems using time and product decompositions are formed using the logic presented in Helber and Sahling (2010). We present the same logic here with the additional notation given in the following. The sub-problem for a given set $KP^{fix} \subseteq KP$ can be stated as follows (Helber and Sahling, 2010):

Sets:

$(i, t) \in KP$ set of product-period combinations

$KP^{opt} \subseteq KP$ set of product-period combinations for which binary variables Y_{it} and U_{it} are optimised in the current sub-problem

$KP^{fix} \subseteq KP$ set of product-period combinations for which binary variables Y_{it} and U_{it} are fixed in the current sub-problem

Parameters:

$\bar{Y}_{it}$ exogenous value of the fixed setup carryover variables U_{it} in the current sub-problem

$\bar{U}_{it}$ exogenous value of the fixed setup variables Y_{it} in the current sub-problem.

The sub-problem:

Minimise objective function (7) subject to constraints (8) to (18) and the additional constraints.

$$Y_{it} = \bar{Y}_{it} \quad \forall (i,t) \in KP^{fix} \tag{19}$$

$$U_{it} = \bar{U}_{it} \quad \forall (i,t) \in KP^{fix} \tag{20}$$

More information on these decomposition schemes can be found in Goren et al. (2012) and Goren and Tunali (2015).

3.2 Genetic algorithms

The binary variables Y_{it} and U_{it} stated in the mathematical model are used in representing a chromosome in GAs. A chromosome used in GAs is a matrix representation in which rows show the setup (Y_{it}) and columns show the setup carryover variables (U_{it}). So, the length of the chromosome is the number of periods (P) multiplied by the number of products (K). Instead of a pure random initial population, the initialisation scheme proposed in Goren et al. (2012) is used. While some part of the initial population is generated randomly, the other part is generated based on problem specific information, in a smart way. The smart way starts with solving the LP relaxation of the CLSP which is the basis of the problem considered in this study. Then using this information, the first parts of chromosomes, i.e., the setups, are generated. For the second part where the setup carryovers are located, a heuristic has been proposed to create the initial values. These values are generated considering the values of the setup variables and setup costs of the products. For more details about this initialisation scheme the reader can refer to Goren et al. (2012) and Goren and Tunali (2015). Similar to our previous work in Goren et al. (2012), the half of the initial population is created randomly while the other half is created using problem specific information.

After creating the initial population and calculating the fitness for all chromosomes, modified roulette wheel selection operator is used in selecting the parents (Goren et al., 2012). To produce offsprings, single point crossover is used. After crossover, single bit flip mutation operator is applied. However, crossover and mutation operators might lead to some infeasible chromosomes in the population. For infeasible chromosomes, a number of repair operators are proposed based on the type of infeasibilities. If the infeasibility originates from capacity violation, these infeasible chromosomes are penalised. To keep the best individual in the population for the next generation, the elitist strategy is used. As mentioned before, in this study, GAs are combined with FOPT heuristic in two different ways to form the hybrid approaches given in Table 1.

Table 1 Summary of the proposed hybrid approaches

		Definition of the hybrid approach
Sequential hybrid approaches	Hybrid Approach 1 (H1)	GAs first and then the fix-and-optimise heuristic with *time decomposition* only.
	Hybrid Approach 2 (H2)	GAs first and then the fix-and-optimise heuristic with *product decomposition* only.
	Hybrid Approach 3 (H3)	GAs first and then the fix-and-optimise heuristic with *time decomposition* first, then *product decomposition.*
	Hybrid Approach 4 (H4)	GAs first and then the fix-and-optimise heuristic with *product decomposition* first, then *time decomposition*.
Embedded hybrid approaches	Hybrid Approach 5 (H5)	The fix-and-optimise heuristic with *time decomposition* in the loop of GAs.
	Hybrid Approach 6 (H6)	The fix-and-optimise heuristic with *product decomposition* in the loop of GAs.
	Hybrid Approach 7 (H7)	The fix-and-optimise heuristic with *time decomposition* in one generation and *product decomposition* in another generation in the loop of GAs.
	Hybrid Approach 8 (H8)	The fix-and-optimise heuristic with two decomposition schemes in the sequence of *product* and *time decomposition* in the loop of GAs.

4 Computational results

This section contains three sets of experiments. Firstly, a set of experiments is conducted to select a criterion for the implementation of the proposed product decomposition scheme. Next, the performances of the proposed hybrid approaches are analysed with respect to the average gaps obtained from a commercial solver (CPLEX) and pure GAs (PGA). Lastly, the sensitivity of the two best performing approaches to changes in problem parameters such as backorder costs, setup times, demand variability, etc., is investigated.

4.1 Analysis of the proposed hybrid approaches

The computational experiments were carried out on the test instances given in Trigeiro et al. (1989) which were modified by multiplying the given demand levels by 1.1 so that excess demand can be backordered (see Table 2). The backorder cost was taken as a linear function of the holding cost ($b = fh$), where $f = 2$ as in Millar and Yang (1984).

Table 2 Classification of data instances

	Problem class					
	1	*2*	*3*	*4*	*5*	*6*
Number of products	6	12	24	6	12	24
Number of periods	15	15	15	30	30	30
Number of instances	5	5	5	5	5	5

The performances of the proposed hybrid approaches were compared to the lower bounds obtained from the simple plant location (SPL) formulation of the MIP model (Stadtler, 1996; Suerie and Stadtler, 2003). The solution quality of the proposed approaches was measured by computing the gap as follows.

$$Gap = 100 * \frac{(heuristic\ solution - lower\ bound)}{lower\ bound} \tag{21}$$

Preliminary experiments were performed to determine the run-time for the FOPT heuristic and it was observed that limiting this value to two seconds helped to keep the overall computational time at a reasonable level. All computations were carried out on a PC with Dual Core, 2 GHz microprocessor and 2 GB RAM. The pure and hybrid GAs were coded in Visual C++ 2008 Express Edition and all sub-problems were solved using Concert Technology of Cplex 11.2. In order to improve the performance of GAs, it is important to select efficient control parameters. Some experiments were carried out on a medium size problem with ten products and 20 periods. Based on these experiments, population size, crossover rate and mutation rate were set to 100, 0.95 and 0.005 for 100 generations.

As mentioned in section three, the proposed product decomposition scheme uses five types of product specific criteria in forming the sub-problems. To decide whether the sub-problems should be formed in ascending, descending or random ordering of these five criteria a pilot experiment was carried out and noted that the performance of the proposed product decomposition scheme was not substantially affected by the order of these criteria. However, since descending order of the criteria resulted in slightly better performance than the other two, in the following computational experiments, descending order of these criteria was used. To evaluate the performance of H2 (GA+product decomposition), ten independent runs were carried out using descending order of these five criteria. Experimental results given in Table 3 are based on the average of these runs.

Table 3 Average% gap over lower bound

	According to 'setup cost'	*According to 'holding cost'*	*According to '(setup cost/ holding cost)'*	*According to 'total demand'*	*According to the 'total cost'*
Class 1	27.7%	28.07%	27.6%	28.14%	*27.4%*
Class 2	*34.3%*	34.5%	34.5%	34.5%	34.5%
Class 3	*15.1%*	15.4%	15.3%	15.3%	15.2%
Class 4	*14.2%*	14.8%	14.6%	14.6%	14.6%
Class 5	6.4%	6.5%	*6.2%*	6.3%	6.3%
Class 6	*6.8%*	7.04%	7%	6.9%	7.03%

Table 4 Summary of experimental studies (based on average of ten runs)

	CPLEX	*PGA*	*Proposed hybrid approaches*							
			Sequential hybrid approaches				*Embedded hybrid approaches*			
			H1	*H2*	*H3*	*H4*	*H5*	*H6*	*H7*	*H8*
Class 1	*16.57%*	30.16%	21.10%	27.70%	21.05%	21.07%	19.93%	22.57%	20.05%	*19.80%*
Class 2	*8.45%*	34.98%	26.05%	34.27%	26.10%	25.95%	25.80%	29.09%	25.81%	*25.67%*
Class 3	*3.067%*	17.25%	10.10%	15.14%	9.912%	*8.69%*	9.65%	11.53%	9.77%	9.46%
Class 4	26.257%	16.80%	10.57%	14.24%	10.36%	*10.07%*	11.18%	12.29%	11.55%	11.11%
Class 5	10.92%	8.64%	5.57%	6.40%	4.58%	*4.57%*	5.06%	5.26%	4.80%	4.58%
Class 6	5.74%	8.82%	6.86%	6.81%	5.67%	*5.60%*	6.91%	6.30%	6.51%	6.19%

As seen in Table 3, for the majority of the problem classes, the product decomposition based on setup cost resulted in a better performance than the others. Hence, we decided to employ this decomposition criterion in the following computational comparative studies.

Table 4 summarises the percentages of the average gaps obtained by each approach for each problem class. It should be noted that CPLEX, PGA and each proposed approach were run under the same computational times in order to obtain compatible results. The results are given in Table 4.

As seen in the table, for all problem classes the performance of the PGA is the worst and hybridising GAs with FOPT heuristic improves the performance of all proposed hybrid approaches. As mentioned before, these hybrid approaches combine GAs and FOPT in various ways. In the sequential hybrid approaches, the FOPT heuristic uses the final solution of the GAs as the initial solution. In doing so we hope that GA will find a promising region in the search space that will be further explored by the FOPT heuristic to find the best solution. In other four hybrid approaches, we used the FOPT heuristic as a diversification strategy by embedding it into the loop of GA. Based on the results stated in Table 4, we can state that sequential hybrid approaches outperform embedded hybrid approaches in four problem classes out of six. Moreover, we also noticed that H2 and H6 which use only product decomposition scheme have the worst performance among the eight hybrid approaches proposed. The performance of CPLEX is superior to the proposed approaches in classes 1, 2 and 3. On the other hand, all proposed hybrid approaches (i.e., sequential and embedded) outperform CPLEX in problem classes 4 and 5. The deviations obtained by the proposed hybrid approaches are approximately half of the deviations obtained by CPLEX. In class 6, which is the largest problem size, H3, H4 and CPLEX have similar performances but performance of H4 is slightly better than CPLEX. Overall, we can state that H4 (a sequential hybrid approach) and H8 (embedded hybrid approach) have better performance than others. As it can be seen from Table 1, the common feature of these approaches is that they both use first product decomposition scheme then time decomposition scheme.

Next, we investigated whether the differences between H4 and H8 were statistically significant at 0.05 level. The results of paired t-test for each problem class are presented in Table 5.

Table 5 Results of the paired-t test to compare H4 and H8

Class	*Paired-t test*			
	X_{H4}–X_{H8}	*Half-length*	*Interval*	*p-value*
1	0.0126	0.17	(–0.157, 0.183)	0.847
2	0.004	0.0068	(–0.0028, 0.0108)	0.178
3	0.00348	0.0071	(0.003637, 0.010597)	0.246
4	–0.0284	0.04	(–0.0684, 0.01157)	0.120
5	–0.0004	0.00258	(–0.002975, 0.002175)	0.688
6	–0.016	0.0288	(–0.0448, 0.0128)	0.197

Based on these results, we can state that there are no statistically significant differences between H4 and H8 (i.e., all *p*-values are greater than 0.05).

4.2 Sensitivity analysis

The primary purpose of sensitivity analysis is to gain insight on how the performances of the best performing approaches (H4 and H8) are affected by different levels of backorder cost. To generate test problems, we redefined the problem with 10 products and 20 periods in Trigeiro et al. (1989) by including backordering cost which was defined as a linear function of the holding cost (b = fh), where f> = 1 as in Millar and Yang (1994). Besides backorder cost, Table 6 summarises other experimental factors used in generating the problem instances: capacity utilisation, time between orders (TBO), demand variability, and setup time. The TBO is changed by changing the setup cost. It should be noted that demand variability is represented by coefficient of variation of demand and the holding cost is assumed to be constant for all products.

Table 6 The experimental design

Parameter	*Values used*	*Total values*
Backorder cost	2 * Holding cost (low)	2
	6 * Holding cost (high)	
Capacity utilisation	75% (low)	2
	85% (high)	
Time between orders (TBO)	1 Period (low)	3
	2 Periods (medium)	
	4 Periods (high)	
Setup time	11 Units of capacity (low)	2
	43 Units of capacity (high)	
CV of demand	0.35 (low)	2
	0.59 (high)	
	Total parameter combinations	48
	Number of problems/combinations	5
	Total problems	240

For each combination, five random problems are generated. Thus, a total of 240 problems are generated to test the performances of the proposed approaches H4 and H8. Experimental results are summarised in Table 7 and also in Figures 3 and 4 which plot the gap for each instance, average gap of five instances at each design point and overall average gap for both H4 and H8.

Below, we summarise which proposed approach will be better than others under different problem-specific parameters:

- As can be seen from the table and also from figures, 3 and 4 as compared to low backorder cost (b = 2h, design points 1 through 24), high backorder cost (b = 6h, design points 25 through 48) results in lower average gap and overall, the performance of H8 is slightly better than H4 both for low and high backorder cost.
- The performance of both algorithms deteriorates when TBO is set to high values and vice versa at low values of TBO. Moreover, performance of H4 (see design points 9 through 12 and 21 through 24) is slightly worse than H8 (see design points

33 through 36 and 45 through 48) for high values of TBO. TBO is related to the setup cost as defined in Trigeiro et al. (1989). Since objective is to minimise total cost, as expected, high setup cost (i.e., TBO is high) may lead to conservation of the setup state for consecutive periods instead of placing orders frequently.

- As far as the capacity utilisation, there are minor differences in the average gaps for both levels of backorder costs. Millar and Yang (1994) state that for a special case of the CLSP with backordering the performance of the algorithms gets worse as the capacity becomes tighter. As can be seen from Table 7, the gaps get worse as the capacity becomes tighter and larger gaps were observed when TBO and capacity utilisation were set to high values.
- Regardless of the value of backorder cost, the performances of both H4 and H8 get better as variability of demand increases. Therefore, we can state that both approaches are cost-effective when CV of demand is high.

Figure 3 Experimental results for the proposed approach H4 (see online version for colours)

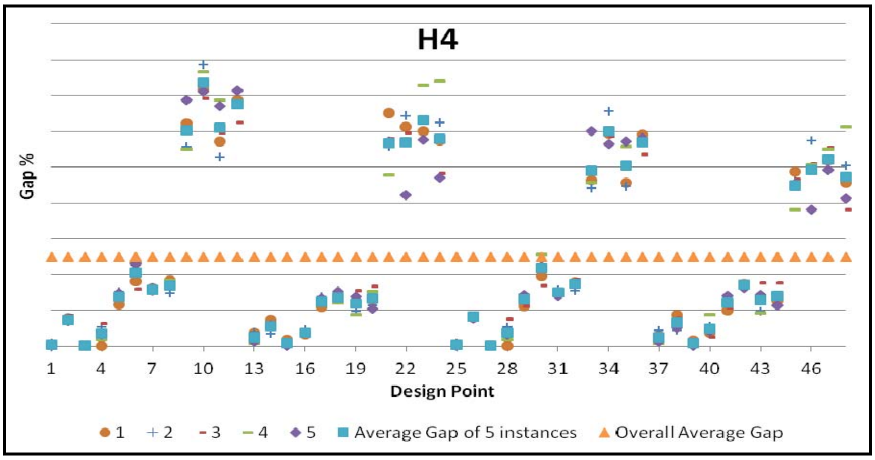


Figure 4 Experimental results for the proposed approach H8 (see online version for colours)

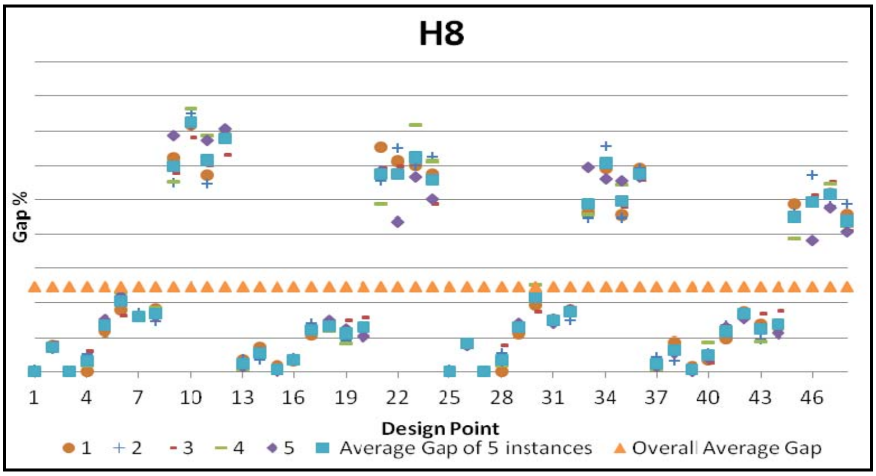


Table 7 Average gaps under different experimental factors for proposed approaches H4 and H8

Setting no.	*Backorder cost*	*CV in demand*	*TBO*	*Setup time*	*Capacity utilisation*	*H4 overall average gap*	*H8 overall average gap*	*Setting no.*	*Backorder cost*	*CV in demand*	*TBO*	*Setup time*	*Capacity utilisation*	*H4 overall average gap*	*H8 overall average gap*
1	Low	Low	Low	Low	Low	0.10%	0.10%	25	High	Low	Low	Low	Low	0.10%	0.10%
2	Low	Low	Low	Low	High	3.66%	3.64%	26	High	Low	Low	Low	High	4.07%	4.09%
3	Low	Low	Low	High	Low	0.00%	0.00%	27	High	Low	Low	High	Low	0.00%	0.00%
4	Low	Low	Low	High	High	1.67%	1.63%	28	High	Low	Low	High	High	1.80%	1.81%
5	Low	Low	Medium	Low	Low	6.86%	6.76%	29	High	Low	Medium	Low	Low	6.48%	6.47%
6	Low	Low	Medium	Low	High	10.31%	10.21%	30	High	Low	Medium	Low	High	10.83%	10.85%
7	Low	Low	Medium	High	Low	7.91%	8.02%	31	High	Low	Medium	High	Low	7.50%	7.36%
8	Low	Low	Medium	High	High	8.51%	8.39%	32	High	Low	Medium	High	High	8.69%	8.77%
9	Low	Low	High	Low	Low	30.15%	29.88%	33	High	Low	High	Low	Low	24.47%	24.34%
10	Low	Low	High	Low	High	36.77%	36.32%	34	High	Low	High	Low	High	30.08%	30.27%
11	Low	Low	High	High	Low	30.53%	30.76%	35	High	Low	High	High	Low	25.27%	24.82%
12	Low	Low	High	High	High	33.74%	33.88%	36	High	Low	High	High	High	28.48%	28.82%
13	Low	High	Low	Low	Low	1.24%	1.14%	37	High	High	Low	Low	Low	1.21%	1.12%
14	Low	High	Low	Low	High	2.80%	2.79%	38	High	High	Low	Low	High	3.31%	3.16%
15	Low	High	Low	High	Low	0.37%	0.37%	39	High	High	Low	High	Low	0.33%	0.33%
16	Low	High	Low	High	High	1.81%	1.73%	40	High	High	Low	High	High	2.45%	2.43%
17	Low	High	Medium	Low	Low	6.12%	6.04%	41	High	High	Medium	Low	Low	6.06%	5.89%
18	Low	High	Medium	Low	High	6.74%	6.64%	42	High	High	Medium	Low	High	8.60%	8.41%
19	Low	High	Medium	High	Low	5.95%	5.64%	43	High	High	Medium	High	Low	6.44%	6.18%
20	Low	High	Medium	High	High	6.66%	6.39%	44	High	High	Medium	High	High	7.02%	6.92%
21	Low	High	High	Low	Low	28.38%	28.70%	45	High	High	High	Low	Low	22.50%	22.46%
22	Low	High	High	Low	High	28.43%	28.78%	46	High	High	High	Low	High	24.65%	24.57%
23	Low	High	High	High	Low	31.62%	31.13%	47	High	High	High	High	Low	26.05%	25.77%
24	Low	High	High	High	High	28.97%	27.97%	48	High	High	High	High	High	23.69%	21.89%
Avg. gap						13.30%	*13.20%*	Avg. gap						11.67%	*11.53%*

Furthermore, we investigated the effects of interaction between different experimental factors on the performances of H4 and H8 under two levels of backorder costs. As can be seen from Tables 8 through 14, both H4 and H8 show similar performance for different levels of backorder cost. Findings from this analysis are:

- Across two levels of setup time and for low level of TBO, both H4 and H8 perform much better than medium and high levels of TBO (see Table 8) and overall, the performance of H8 is slightly better than H4. Smallest average gaps were observed for high setup times and low TBO but average gaps were largest when both setup time and TBO were set to high values. It seems that there is a slight negative effect of high backorder cost on performance of both approaches.
- Regarding the interaction between TBO and demand variability (see Table 9), across two levels of coefficient of variation of demand, smallest average gaps were observed for low values of TBO. Moreover, while the effect of high backorder on performances of two approaches was slightly negative (i.e., larger average gaps) for low and medium levels of TBO, smaller average gaps were observed for high backorder cost and for high TBO.
- The simultaneous effect of TBO and capacity utilisation is presented in Table 10. As can be seen from the table, across two levels of capacity both approaches perform better for low levels of TBO and average gaps increase for high levels of TBO. However, negative effect of high TBO on performances of both approaches seems to decrease when backorder cost is set to six times of holding cost.
- Under high demand variability, H4 and H8 perform better across all levels of setup times and high backorder cost positively affects (i.e., lower average gaps) the performances of both approaches (see Table 11).
- As can be seen from Table 12 that both approaches perform better for low capacity cases across all levels of setup time and high backorder costs further improve the performances of both approaches (i.e., lower average gaps are observed).
- The simultaneous effect of demand variability and capacity utilisation on the gap is shown in Table 13. Across all levels of capacity, H4 and H8 perform better under high demand variability and as stated above, the performances of both approaches further improve for high backorder cost.

Table 8 The effects of interaction of TBO and setup time on gap

	Setup time	*H4*			*H8*		
		TBO			*TBO*		
		Low	*Medium*	*High*	*Low*	*Medium*	*High*
$b = 2h$	Low	1.95%	7.51%	30.93%	1.90%	7.43%	30.88%
	High	0.96%	7.26%	31.21%	0.93%	7.09%	30.85%
$b = 6h$	Low	2.17%	7.99%	25.42%	2.099%	7.89%	25.43%
	High	1.14%	7.41%	25.87%	1.12%	7.31%	25.25%

Table 9 The effect of interaction of TBO and CV on gap

	CV	H4			H8		
		TBO			TBO		
		Low	Medium	High	Low	Medium	High
$b = 2h$	Low	1.36%	8.40%	32.80%	1.33%	8.35%	32.71%
	High	1.56%	6.37%	29.35%	1.49%	6.17%	29.03%
$b = 6h$	Low	1.49%	8.37%	27.08%	1.50%	8.37%	27.07%
	High	1.82%	7.03%	24.22%	1.72%	6.83%	23.61%

Table 10 The effect of interaction of TBO and capacity utilisation on gap

	Capacity utilisation	H4			H8		
		TBO			TBO		
		Low	Medium	High	Low	Medium	High
$b = 2h$	Low	0.43%	6.71%	30.17%	0.39%	6.61%	29.95%
	High	2.48%	8.05%	31.97%	2.44%	7.91%	31.79%
$b = 6h$	Low	0.41%	6.62%	24.57%	0.39%	6.47%	24.27%
	High	2.90%	8.79%	26.73%	2.83%	8.73%	26.41%

Table 11 The effect of interaction of setup time and CV on gap

	Capacity utilisation	H4		H8	
		Setup time		Setup time	
		Low	High	Low	High
$b = 2h$	Low	14.64%	13.73%	14.47%	13.80%
	High	12.29%	12.56%	12.34%	12.12%
$b = 6h$	Low	12.67%	11.96%	12.70%	11.93%
	High	11.06%	11.00%	10.92%	10.53%

Table 12 The effect of interaction of setup time and capacity utilisation on gap

	Capacity utilisation	H4		H8	
		Setup time		Setup time	
		Low	High	Low	High
$b = 2h$	Low	12.14%	12.73%	12.07%	12.57%
	High	14.78%	13.56%	14.74%	13.35%
$b = 6h$	Low	10.14%	10.93%	10.07%	10.68%
	High	13.59%	12.02%	13.54%	11.78%

Table 13 The effect of interaction of CV and capacity utilisation on gap

	CV	*H4*		*H8*	
		Capacity utilisation		*Capacity utilisation*	
		Low	*High*	*Low*	*High*
$b = 2h$	Low	12.59%	15.78%	12.57%	15.70%
	High	12.28%	12.57%	12.07%	12.40%
$b = 6h$	Low	10.64%	13.99%	10.51%	14.12%
	High	10.43%	11.62%	10.25%	11.20%

Overall, we can state that regardless of the level of backorder cost, proposed hybrid approaches, H4 and H8 perform better when the demand variability is high. Other experimental findings are that the performances of both approaches are better under low setup costs and low capacity utilisation and both approaches have better performances under high setup times as compared to low setup times. Lastly, regarding the effect of different levels of backorder cost on performances of proposed approaches, except for the simultaneous effect between setup time and TBO where average gaps slightly increase for high backorder cost, in all other cases, we observe that high backorder cost helps to further reduce the average gap for high values of TBO and demand variability.

5 Conclusions

Lot sizing is one of the most important problems in production planning. Most of the studies in this area focus on addressing the solution of the CLSP which is known to be NP-Hard. Adding the setup carryover and backordering makes the problem more complex. Therefore, exact solution approaches are not capable of finding good solutions in reasonable computational time. In this study, a number of hybrid meta-heuristic approaches are proposed to deal with the CLSP$^+$. These hybrid approaches combine GAs and FOPT heuristic. To compare the performance of these hybrid approaches, an extensive computational study is carried out. Based on the results of this study, we could state that the performance of GAs can be improved by hybridising it with FOPT heuristic in different ways. We further noticed that the performance of proposed hybrid approaches is affected by the type of the decomposition scheme employed in the FOPT heuristic. Lastly, best performing hybrid approaches are further analysed to investigate how their performances are affected by changes in problem specific parameter values including backorder costs, setup times, setup costs, capacity utilisation and demand variability.

This study considers the CLSP on a single machine with setup carryover and backordering. In a future study, the problem studied in this paper can be extended to include other features such as parallel machines, lost sales and sequence dependent setup times and costs and the proposed hybrid approaches can be modified to deal with these new features. A recent trend in this area is to present solution approaches for the multi-objective lot sizing problems including lot sizing, distribution and scheduling decisions. Another interesting research direction is to investigate the performances of the proposed approaches presented in this paper on this kind of problems.

References

Almada-Lobo, B., Klabjan, D., Carravilla, M.A. and Oliveira, J.F. (2007) 'Single machine multi-product capacitated lot sizing with sequence-dependent setups', *International Journal of Production Research*, Vol. 45, No. 20, pp.4873–4894.

Bitran, G. and Yanasse, H.H. (1982) 'Computational complexity of the capacitated lot size problem', *Management Science*, Vol. 28, No. 10, pp.1174–1186.

Degraeve, Z. and Jans, R. (2007) 'A new Dantzig-Wolfe reformulation and Branchand-price algorithm for the capacitated lot sizing problem with set up times', *Operations Research*, Vol. 55, No. 5, pp.909–920.

Florian, M., Lenstra, J.K. and Rinnooy Kan, A.H.G. (1980) 'Deterministic production planning: algorithms and complexity', *Management Science*, Vol. 26, No. 7, pp.669–679.

Gopalakrishnan, M., Ding, K., Bourjolly, J-M. and Mohan, S. (2001) 'A tabu search heuristic for the capacitated lot-sizing problem with set-up carryover', *Management Science*, Vol. 47, No. 6, pp.851–863.

Goren, H.G. and Tunali. S. (2015) 'Solving the capacitated lot sizing problem with setup carryover using a new sequential hybrid approach', *Applied Intelligence*, Vol. 42, No. 6, pp.805–816.

Goren, H.G., Tunali, S. and Jans, R. (2010) 'A review of applications of genetic algorithms in lot sizing', *Journal of Intelligent Manufacturing*, Vol. 21, No. 4, pp.575–590.

Goren, H.G., Tunali, S. and Jans, R. (2012) 'A hybrid approach for the capacitated lot sizing problem with setup carryover', *International Journal of Production Research*, Vol. 50, No. 6, pp.1582–1597.

Helber, S. and Sahling, F. (2010) 'A fix-and-optimize approach for the multi-level capacitated lot sizing problem', *International Journal of Production Economics*, Vol. 123, No. 2, pp.247–256.

Hindi, K.S., Fleszar, K. and Charalambous, C. (2003) 'An effective heuristic for the CLSP with setup times', *The Journal of the Operational Research Society*, Vol. 54, No. 5, pp.490–498.

Jans, R. and Degraeve, Z. (2004) 'Improved lower bounds for the capacitated lot sizing problem with set up times', *Operations Research Letters*, Vol. 32, No. 2, pp.185–195.

Karimi, B. and Ghomi, F. (2002) 'A new heuristic for the CLSP problem with backlogging and set-up carryover', *International Journal of Advanced Manufacturing Systems*, Vol. 5, No. 2, pp.66–77.

Karimi, B., Ghomi, F. and Wilson, J.M. (2006) 'A tabu search heuristic for the CLSP with backlogging and set-up carryover', *Journal of the Operational Research Society*, Vol. 57, No. 2, pp.140–157.

Lang, J.C. and Shen, Z-J.M. (2011) 'Fix-and-optimize heuristics for capacitated lot-sizing with sequence-dependent setups and substitutions', *European Journal of Operation Research*, Vol. 3, p.214.

Manne, A.S. (1958) 'Programming of economic lot sizes', *Management Science*, Vol. 4, No. 2, pp.115–135.

Millar, H.H. and Yang, M. (1994) 'Lagrangean heuristics for the capacitated multi-item lot sizing problem with backordering', *International Journal of Production Economics*, Vol. 34, No. 1, pp.1–15.

Pochet, Y. and Wolsey, L.A. (2000) *Production Planning by Mixed Integer Programming*, Springer, New York, NY.

Quadt. D., Kuhn. H. (2009) 'Capacitated lot-sizing and scheduling with parallel machines. Back-orders. and setup carry-over', *Naval Research Logistics*, Vol. 56, No. 4, pp.366–384.

Sahling, F., Buschkühl, L., Tempelmeier, H. and Helber, S. (2009) 'Solving a multi-level capacitated lot sizing problem with multi-period setup carry-over via a fix-and-optimize heuristic', *Computers and Operations Research*, Vol. 36, No. 9, pp.2546–2553.

Seanner, F., Almada-Lobo, B. and Meyr, H. (2013) 'Combining the principles of variable neighborhood decomposition and the fix-and-optimize heuristic to solve the multi-level lot-sizing and scheduling problems', *Computers and Operations Research*, Vol. 40, pp.303–317.

Sox, C.R. and Gao, Y. (1999) 'The capacitated lot-sizing problem with set-up carryover', *IIE Transactions*, Vol. 31, No. 2, pp.173–181.

Stadtler, H. (1996) 'Mixed integer programming model formulations for dynamic multi-item multi-level capacitated lot sizing', *European Journal of Operations Research*, Vol. 94, No. 3, pp.561–581.

Suerie, C. and Stadtler, H. (2003) 'The capacitated lot-sizing problem with linked lot sizes', *Management Science*, Vol. 49, No. 8, pp.1039–1054.

Toledo, C.F.M., de Oliveira, R.R.R. and França, P.M. (2013) 'A hybrid multi-population genetic algorithm applied to solve the multi-level capacitated lot sizing problem with backlogging', *Computers and Operations Research*, Vol. 40, No. 4, pp.910–919.

Trigeiro, W., Thomas, L.J. and McClain, J.O. (1989) 'Capacitated lot sizing with setup times', *Management Science*, Vol. 35, No. 3, pp.353–366.

Wagner, H.M. and Whitin, T.M. (1958) 'Dynamic version of the economic lot size model', *Management Science*, Vol. 5, No. 1, pp.89–96.

Xiao, J., Zhang, C., Zheng, L. and Gupta, J.N.D. (2013) 'MIP-based fix-and-optimise algorithms for the parallel machine capacitated lot-sizing and scheduling problem', *International Journal of Production Research*, Vol. 51, No. 16, pp.5011–5028.